

## Metadata

## Transcript

```
--DIFF--  
diff --git a/tests/test_provider_gemini.py b/tests/test_provider_gemini.py  
index 141af81..5df2ef4 100644  
--- a/tests/test_provider_gemini.py  
+++ b/tests/test_provider_gemini.py  
@@ -101,3 +101,107 @@ def test_gemini_provider_upload_fail():  
     provider = GeminiProvider(api_key="dummy_api_key", model_name="gemini-2.5-flash")  
     segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt")  
     assert len(segments) == 0  
  
+  
+  
+  
+# --- 01/P11: Gemini remote-file GC (startup orphan sweep) ---  
+  
+def _file_mock(name, mime, create_time_iso, state="ACTIVE"):  
+    f = MagicMock()  
+    f.name = name  
+    f.mime_type = mime  
+    f.create_time = create_time_iso  
+    f.state = MagicMock()  
+    f.state.name = state
```

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+     return f
+
+
+
+def test_cleanup_orphaned_files_deletes_stale_videos_only():
+    """Deletes video files older than the grace window; leaves non-videos and fresh files."""
+    from datetime import datetime, timedelta, timezone
+    stale = (datetime.now(timezone.utc) - timedelta(hours=2)).isoformat()
+    fresh = (datetime.now(timezone.utc) - timedelta(minutes=5)).isoformat()
+    files = [
+        _file_mock("old-vid", "video/mp4", stale),          # stale video -> deleted
+        _file_mock("fresh-vid", "video/quicktime", fresh),   # fresh video -> kept (grace)
+        _file_mock("old-img", "image/png", stale),           # non-video -> kept
+        _file_mock("notts", "video/mp4", None),              # no timestamp -> kept (safe)
+    ]
+    with patch("backend.providers.gemini.genai"):
+        provider = GeminiProvider(api_key="k", model_name="m")
+        provider._client = MagicMock()
+        provider._client.files.list.return_value = iter(files)
+        deleted = provider.cleanup_orphaned_files(grace_sec=3600)
+    assert deleted == 1
+    provider._client.files.delete.assert_called_once_with(name="old-vid")
+
+
+def test_cleanup_orphaned_files_list_failure_is_swallowed():
+    """A list() failure must not raise ? the sweep is best-effort and must not block startup."""
+    with patch("backend.providers.gemini.genai"):
+        provider = GeminiProvider(api_key="k", model_name="m")
+        provider._client = MagicMock()
+        provider._client.files.list.side_effect = RuntimeError("network")
+        assert provider.cleanup_orphaned_files() == 0
+
+
+def test_cleanup_orphaned_files_per_file_delete_failure_is_swallowed():
+    """A delete() failure on one file must not abort the sweep."""
+    from datetime import datetime, timedelta, timezone
+    stale = (datetime.now(timezone.utc) - timedelta(hours=2)).isoformat()
+    files = [
+        _file_mock("a", "video/mp4", stale),
+        _file_mock("b", "video/mp4", stale),
+    ]
+    with patch("backend.providers.gemini.genai"):
+        provider = GeminiProvider(api_key="k", model_name="m")
+        provider._client = MagicMock()
+        provider._client.files.list.return_value = iter(files)
+        # First delete raises, second succeeds ? _safe_delete swallows it.
+        provider._client.files.delete.side_effect = [RuntimeError("boom"), None]
+        deleted = provider.cleanup_orphaned_files(grace_sec=3600)
+    assert deleted == 1 # only the second file counted
+
+
+# --- 01/P11: main.py startup wiring (_maybe_sweep_gemini_orphans) ---
+
+def test_maybe_sweep_skips_non_gemini_provider(monkeypatch):
+    """A non-Gemini provider must short-circuit (no client built, no sweep)."""
+    import backend.main as main_mod
+    cfg = MagicMock()
+    cfg.ai_provider = "openai"
+    called = []
+    monkeypatch.setattr(main_mod.provider_registry, "get",
+                        lambda *a, **k: called.append(k) or MagicMock())
+    monkeypatch.setenv("GEMINI_API_KEY", "k")
+    main_mod._maybe_sweep_gemini_orphans(cfg)
+    assert called == [] # provider never instantiated

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+
+
+def test_maybe_sweep_skips_when_no_api_key(monkeypatch):
+    """No GEMINI_API_KEY -> no sweep (can't auth)."""
+    import backend.main as main_mod
+    cfg = MagicMock()
+    cfg.ai_provider = "gemini"
+    monkeypatch.delenv("GEMINI_API_KEY", raising=False)
+    called = []
+    monkeypatch.setattr(main_mod.provider_registry, "get",
+                        lambda *a, **k: called.append(k) or MagicMock())
+    main_mod._maybe_sweep_gemini_orphans(cfg)
+    assert called == []
+
+
+def test_maybe_sweep_runs_and_swallows_errors(monkeypatch):
+    """Gemini + key present -> sweep runs; a provider error is swallowed (never blocks
+startup)."""
+    import backend.main as main_mod
+    cfg = MagicMock()
+    cfg.ai_provider = "gemini"
+    cfg.ai_model = "gemini-2.5-flash"
+    provider = MagicMock()
+    provider.cleanup_orphaned_files.side_effect = RuntimeError("api down")
+    monkeypatch.setattr(main_mod.provider_registry, "get", lambda *a, **k: provider)
+    monkeypatch.setenv("GEMINI_API_KEY", "k")
+    # Must not raise.
+    main_mod._maybe_sweep_gemini_orphans(cfg)
+    provider.cleanup_orphaned_files.assert_called_once()

```

#### 4. assistant (2026-06-25T16:56:46.200Z)

The diff is clear. Now let me verify the exact blank-line counts at the locations the finding describes.

#### 5. user (2026-06-25T16:56:47.692Z)

```

99:[         client_mock.files.upload.side_effect = Exception("Upload err")]
100:[]
101:[         provider = GeminiProvider(api_key="dummy_api_key", model_name="gemini-2.5-flash")]
102:[         segments, metrics = provider.analyze_video("dummy.mp4", "dummy prompt")]
103:[         assert len(segments) == 0]
104:[]
105:[]
106:[]
107:[# --- 01/P11: Gemini remote-file GC (startup orphan sweep) ---]
108:[]
109:[def _file_mock(name, mime, create_time_iso, state="ACTIVE"):]
110:[     f = MagicMock()]
111:[     f.name = name]
112:[     f.mime_type = mime]

```

#### 6. assistant (2026-06-25T16:56:49.993Z)

There are three blank lines (104, 105, 106) before the first new GC section. Let me check the second location (before the main.py-wiring section).

#### 7. user (2026-06-25T16:56:51.475Z)

```

107:[# --- 01/P11: Gemini remote-file GC (startup orphan sweep) ---]
167:[# --- 01/P11: main.py startup wiring (_maybe_sweep_gemini_orphans) ---]
---around main wiring section---
145:[         assert provider.cleanup_orphaned_files() == 0]
146:[]

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147:[]
148:[def test_cleanup_orphaned_files_per_file_delete_failure_is_swallowed():]
149:[    """A delete() failure on one file must not abort the sweep."""]
150:[    from datetime import datetime, timedelta, timezone]
151:[    stale = (datetime.now(timezone.utc) - timedelta(hours=2)).isoformat()]
152:[    files = []]
153:[        _file_mock("a", "video/mp4", stale),]
154:[        _file_mock("b", "video/mp4", stale),]
155:[    ]]
156:[    with patch("backend.providers.gemini.genai"):]

```

#### 8. user (2026-06-25T16:56:55.482Z)

```

163:[    assert deleted == 1 # only the second file counted]
164:[]
165:[]
166:[]
167:[# --- 01/P11: main.py startup wiring (_maybe_sweep_gemini_orphans) ---]
168:[]
169:[def test_maybe_sweep_skips_non_gemini_provider(monkeypatch):]
170:[    """A non-Gemini provider must short-circuit (no client built, no sweep)."""]

```

#### 9. assistant (2026-06-25T16:56:59.952Z)

Both locations confirmed. Lines 104-106 (three blank lines before the GC section) and lines 164-166 (three blank lines before the main-wiring section). Let me confirm ruff is genuinely clean on this (the finding claims CI is green and it's not a ruff failure).

#### 10. user (2026-06-25T16:57:01.209Z)

All checks passed!  
EXIT: 0

#### 11. assistant (2026-06-25T16:57:02.990Z)

Verification complete. The finding is factually accurate on every point.

#### 12. user (2026-06-25T16:57:08.026Z)

Structured output provided successfully

#### 13. assistant (2026-06-25T16:57:10.933Z)

The finding is verified as real and correctly classified as a NIT. Both cited locations in `tests/test\_provider\_gemini.py` (lines 104-106 and 164-166) contain triple blank lines before the new section comment headers, ruff passes clean, and it's purely cosmetic with no functional impact.